

Research Statement

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Capacity-constrained service systems such as primary care, ambulance response, organ allocation, sales territories, last-mile delivery face the same operational question that is centered around how service-region boundaries might be drawn and experts might be placed so that there would be no community that is systematically underserved. My main doctoral research, *FalCom*, develops a single sampling framework that answers this question across application domains, treating it as a distributional question over the space of feasible plans rather than a search for a single optimum. FalCom is the first Markov chain Monte Carlo method that samples ensembles of feasible service-region designs at city or national scale, partitioning a geographic region into contiguous, capacity-demand balanced districts at multiple administrative levels and assigning service teams within each district. It uses advanced combinatorial algorithms and delivers empirical performance an order of magnitude beyond published methods, handling 50,000 geographical basic units, and is released as the open-source **FalcomChain** Python library with visualization tools that support decision-making and policy evaluation. FalCom received a research fellowship from SoReMo (Socially Responsible Modeling) for equitable approaches in computation, modeling, and design, and was accepted to ISCO 2026 with a travel grant from IIT.

Current research

Equitable Expansion of FQHCs in Chicago. The primary care safety network of Chicago is largely operated by Federally Qualified Health Centers (FQHCs) and consists of community-based clinics for the low-income and uninsured people that are supported by federal funding from the Health Center Program of the U.S. Health Resources and Services Administration (HRSA), over 32 million patients nationally in 2024. HRSA’s Compliance Manual requires bounding a catchment area so services stay accessible, siting clinics relative to where patients live and work, and maintaining hospital referral relationships, which are unsolved spatial-optimization problems. Formulating them faithfully both exposes the failures of the current designation process and shows how to repair them.

We identify four structural gaps in the current system: (i) HRSA has no quantitative tool for deciding where to site new centers or how to draw their service areas, (ii) the formula designating underserved areas dates from the 1970s, (iii) catchment area designation is manipulable (Kim et al. (Health Equity, 2020) found 87 eligible but undesignated Chicago tracts), and (iv) unmapped referral networks. As infrastructure, I built a spatial inventory integrating six data sources across 166 FQHC sites, 128 hospitals, and 61,460 providers, with a travel-time matrix from every census block to every facility, plus an interactive dashboard. The analytical core has three parts: a hierarchical district-superdistrict design via a flow-based FalCom sampler, a continuous underservice score replacing the binary IMU cutoff, and enabling policy comparison rather than a yes/no verdict. Next steps are full FalCom runs on the calibrated data, public deployment of the decision-support tool, and an NIH application.

Proposed research directions

Direction 1: Stochastic and flow-based extensions of FalCom. The current FalCom framework treats demand as deterministic assignment and capacity as a deterministic team-level workload. I propose two extensions. First, a stochastic demand model: a two-stage formulation that

makes first-stage districting decisions under uncertainty and second-stage operational adjustments to realized demand, with chance-constrained district feasibility, convergence analysis under a stochastic objective, and distributionally robust extensions. Second, a flow-based capacity model within FalCom: rather than treating capacity as a fixed per-team workload, capacity is measured through flow conservation and flow cost across hierarchy levels, for systems where the allocated quantity is referral flow and patient flow rather than local population. This sits within the stochastic-OR tradition spanning healthcare scheduling, capacity planning, supply-chain network design, and congestion modelling.

Direction 2: Equity in Canadian kidney allocation system. In many national-scale service systems the infrastructure is largely fixed. The policy question is whether existing boundaries deliver equitable access and whether specialized teams could be placed differently within the existing infrastructure to close gaps. Indigenous Canadians are 66% less likely to receive a kidney transplant and face longer wait times despite a three-fold higher risk of kidney failure (Gill and Kaur, *Canadian Journal of Kidney Health and Disease*, 2025). This raises an ensemble-comparison question no single-optimum method can answer: how do current donor service area boundaries compare to the distribution of feasible alternatives, and which geographic regions and Indigenous communities are systematically underserved across that distribution? The methodological lineage parallels Gentry et al. (*Interfaces*, 2015), who used districting optimization to redesign U.S. liver allocation, and Akshat, Gentry, and Raghavan (*Health Care Management Science*, 2024) extended this with heterogeneous donor circles at 58 donation service areas. FalCom can extend the line to the 50,000-node scale, and Direction 1 lets the analysis incorporate uncertain demand and supply across provinces. I have been discussing the US case with Dr. Sommer Gentry at NYU’s Grossman School of Medicine. The Canadian case is a natural extension.

Direction 3: Districting and UAV routing for large-scale emergency response. Building on my UAV routing and FalCom work, I plan to integrate hierarchical districting with within-district UAV routing for emergency-response design at the scale and realism required for operational use. At the upper level, FalCom designs response zones on real geographic dual graphs (dissemination areas, census areas, fire-management units) under capacity, contiguity, and equity constraints. At the lower level, the UAV-routing matheuristic plans within-district missions under energy, time-window, and information-collection objectives on a real-world UAV platform. Concrete applications include post-disaster damage mapping in the first 72 hours of a response, wildfire monitoring under decaying information value, and multi-agency search-and-rescue coordination.

Direction 4: Chicago physician referral network. Chicago has one of the largest FQHC networks in the US including major academic medical centers such as UChicago, Northwestern, Rush, and UIC, yet no one has mapped how referrals actually flow between primary care and specialists in the city. An et al. (*Statistics in Medicine*, 2018) constructed the national physician referral network from the CMS Physician Shared Patient Patterns dataset, where physicians are nodes and shared Medicare patients within 30-day windows form edges. The national network contains over one million physicians and 55 million referral records. They found strong hub-and-spoke structure with 93% of referral steps staying within the same hospital region, and referral volumes following a gravity law inversely proportional to distance. CMS made this data publicly available between 2009 and 2015 but has not released it since; obtaining a current release would require a ResDAC application or a CMS data use agreement. The natural follow-on is to overlay the referral network onto the Chicago FQHC network to identify referral deserts where patients have no viable path from primary care to specialist care.